

SIMULATION OF TIME-DIVISION-MULTIPLEXER USING VHDL

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VHDL stands for very high-speed integrated circuits (VH-SIC) hardware description language. It is a programming language used for designing, simulating and synthesising any complex digital circuit. In this article, we describe the design and simulation of a time-division-multiplexer (TDM) for digital signals, which is mainly used in commu-

nications, using VHDL. The program is developed and tested in Xilinx VHDL software under Microsoft's Windows XP environment.

TDM works on the principle of time-sharing between the multiple input lines for a single output line. A known number of signals from input lines are made available periodically on a single output line on recurrent time slots of fixed length. The master clock provides time slots for each

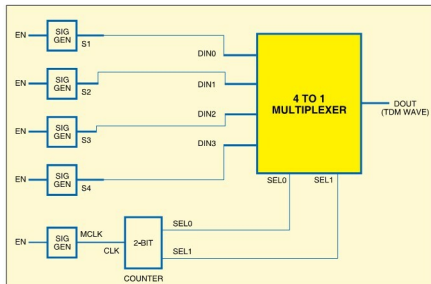


Fig. 1: A time-division-multiplexer

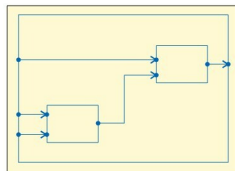


Fig. 2: Entity blocks

of the input signals to be multiplexed. As the master clock needs to accommodate all the input lines that are multiplexed, decision on the time period of the master clock needs to be carefully taken. For this, an arbitrary 'on' and 'off' period for each input signal is selected. The 'on' and 'off' periods of the master clock are derived by adding the time periods of the input signals; for example,

$2 \times (2+4+8+16) = 2 \times 30 \text{ ns}$. The select signals required for multiplexing are derived from the master clock using a 2-bit counter. Fig. 1 describes a simple implementation of a time division multiplexer with four input signals (DIN0 through DIN3) and two select (SEL0 and SEL1) signals. The truth table of this multiplexer is shown below.

Truth Table

Sel inputs		Output
Sel (1)	Sel (0)	dout
0	0	din(0)
0	1	din(1)
1	0	din(2)
1	1	din(3)

The basics of VHDL

The design of digital systems using VHDL has two stages—the first stage of the code describes the external view of the design with a set of input and output signals; the second stage describes the internal view with the detail logic and other functional descriptions.

Before looking into the designing aspects, let us get familiar with VHDL jargons such as entity, architecture, component and component instance.

The entity declaration in the code represents the external interfaces in the design, and the architecture represents the internal functions. An entity represents a microprocessor, hardware module or even a single logic gate. It is represented by a block (rectangular box) shown in Fig. 2. The two blocks inside the larger one are

